A NaMaster bispectrum estimator for the shear–shear–density field: mask, variable noise, and optimal weighting

Abstract

We implement and validate the filtered–squared bispectrum estimator for the shear–shear–density field on realistic simulations with a survey mask and spatially inhomogeneous shape noise. The shear field is inverse-variance weighted (W_1), bandpass-filtered, and squared to form the spin-0 intensity $M_0 = |\tilde{\gamma}|^2$ and the spin-4 field $M_4 = \tilde{\gamma}^2$; these are cross-correlated with the lens density δ through NaMaster, which performs the mode decoupling and supplies a Gaussian covariance. The post-square weight W_2 enters as the NaMaster field weight, so mode-decoupling makes the mean unbiased irrespective of W_2 while W_2 controls the variance. The additive M_0 noise bias is removed with an exact squared-kernel ($|K|^2$) template. Because W_1 is applied before the square (and the mask leaks power across the filter edge), the masked, weighted estimator differs from the full-sky truth by a deterministic transfer function, which we derive from first principles and evaluate four independent ways (Section 3). Using 100 full-sky, noise-free realizations as ground truth and 100×4 disjoint DES Y6 footprints as 400 masked, noisy realizations, we show that (i) the estimator is unbiased given the realization scatter, (ii) the NaMaster Gaussian covariance reproduces the sample scatter, and (iii) optimal weighting improves the signal-to-noise in the noise-dominated bands. The fiducial analysis is inverse-variance weighting ($W_1 = 1/n^2$, W_2 optimal) with the cross-spectra calibrated by an *analytic* transfer function derived from the survey window and the power spectra; it remains unbiased without any calibration on the signal simulations.

1 Estimator and pipeline

The pipeline follows the companion note `optimal_weights.tex` and the `fsb2_pipeline_summary` reference implementation. For a spin-2 shear field $\boldsymbol{\gamma} = (\gamma_1, \gamma_2)$ with inhomogeneous, diagonal shape-noise variance $n^2(\hat{\boldsymbol{n}})$ and a lens overdensity δ, the estimator is

$$\boldsymbol{\gamma} \xrightarrow{W_1} W_1\boldsymbol{\gamma} \xrightarrow{F} \tilde{\boldsymbol{\gamma}} \xrightarrow{(\cdot)^2} M_0, M_4 \xrightarrow{\times\delta} \hat{C}_\ell^{\delta M_0}, \hat{C}_\ell^{\delta M_{4E}}, \tag{1}$$

where F is a sharp top-hat bandpass $f_\ell = \mathbb{1}[\ell \in [\ell_{\rm lo}, \ell_{\rm hi}]]$, $\tilde{\boldsymbol{\gamma}} = F[W_1\boldsymbol{\gamma}]$, and

$$M_0 = \tilde{\gamma}_1^2 + \tilde{\gamma}_2^2, \qquad M_{4E} = \tilde{\gamma}_1^2 - \tilde{\gamma}_2^2, \qquad M_{4B} = 2\tilde{\gamma}_1\tilde{\gamma}_2. \tag{2}$$

The noise model is that of the simulations: source counts trace the source overdensity, so the per-component shape-noise variance is

$$n^2(\hat{\boldsymbol{n}}) = \frac{\sigma_e^2}{\bar{n}\,\Omega_{\rm pix}\,(1+\delta_{\rm src}(\hat{\boldsymbol{n}}))}, \qquad \sigma_e = 0.26,\ \bar{n} = 10\,\mathrm{arcmin}^{-2}, \tag{3}$$

which is largest in voids and is correlated with the lens density.

Weights. We test three schemes:

- **none**: $W_1 = 1$, $W_2 = 1$ (binary mask only);
- **W2**: $W_1 = 1$, W_2 optimal;
- **W1W2** (*fiducial*): $W_1 = 1/n^2$, W_2 optimal.

The fiducial analysis is **W1W2** with the cross-spectra calibrated by the analytic transfer of Section 3; the other schemes and the sim-calibrated transfer are kept for comparison. The optimal post-square weight is the deterministic inverse-variance map of the note,

$$W_2(\hat{\boldsymbol{n}}) \;\propto\; \frac{\mathbb{E}[M_0(\hat{\boldsymbol{n}})]_{\rm sig}}{\mathrm{Var}[M_0(\hat{\boldsymbol{n}})]} = \frac{c\,[\,|K|^2\star(W_1^2)\,](\hat{\boldsymbol{n}})}{\big([\,|K|^2\star(W_1^2(n^2+c))\,](\hat{\boldsymbol{n}})\big)^2}, \tag{4}$$

built from filter-smoothed powers of the noise map via the squared real-space kernel $|K|^2$ (c is the band-averaged per-component signal density). It is a deterministic function of the known noise map and the filter.

Role of NaMaster. W_2 is supplied as the NaMaster *weight map* on the squared field, and δ carries the binary mask. NaMaster's mode-coupling matrix then deconvolves the joint effect of the mask and W_2, so the decoupled bandpowers are unbiased for *any* W_2 shape; W_2 affects only the variance. The overall normalization of W_2 cancels in the decoupling. We use M_0 as a spin-0 field crossed with the spin-0 δ, and (M_{4E}, M_{4B}) as a spin-4 field crossed with δ (the E component is the signal, B a null test).

M_0 **noise bias.** M_0 has a non-zero noise mean $\mathbb{E}[M_0^{\rm noise}] = 2\,\Omega_{\rm pix}\,[\,|K|^2\star(W_1^2 n^2)\,]$, computed exactly with the `bl2beam` squared-kernel route. Because the noise variance tracks $\delta_{\rm src}$, which is correlated with δ, this template has a non-zero cross-correlation with δ and *must* be subtracted; M_4 has zero noise mean (Wick) and needs no debiasing. We verified the template reproduces the Monte-Carlo noise mean of M_0 to $\lesssim 0.5\%$.

The transfer function. $W_1 \neq 1$ is applied *before* the square, so the measured field is $|F[W_1\boldsymbol{\gamma}]|^2$, which differs from the unweighted ground-truth field $|F[\boldsymbol{\gamma}]|^2$ by a deterministic, spatially varying modulation; the pre-filter masking likewise leaks power across the footprint edge. We correct both with a transfer function T_b, reporting results in truth units as $\hat{C}_b^{\rm corr} = \hat{C}_b/T_b$. The derivation of T_b and its four independent evaluations (the measured $T^{\rm (A)}$, the Gaussian Monte-Carlo $T^{\rm (B)}$, the closed form $T^{\rm (C)}$, and the mode-coupling-matrix calibration $T^{\rm (D)}$) are the subject of Section 3; the fiducial result uses the analytic $T^{\rm (B)}$, which removes any dependence on the signal sims.

2 Simulations and configuration

We use 100 realizations of Takahashi-like maps at $N_{\rm side} = 1024$: source shear (γ_1, γ_2) at $z_s \simeq 1.03$, lens density δ at $z_t \simeq 0.54$, and source density $\delta_{\rm src}$ at $z \simeq 1.0$ (for the noise model). The mask is the DES Y6 footprint; the four tomographic footprints `desy6_map1..4` are mutually disjoint ($f_{\rm sky} \simeq 0.113$ each, zero overlap), so applying each to the same sky realization yields 4 quasi-independent samples, for 400 in total.

Filter bands $[\ell_{\rm lo}, \ell_{\rm hi}]$	$[50, 500]$, $[500, 1000]$, $[1000, 1500]$
Cross bandpowers	linear, $\Delta\ell = 48$, analysed to $\ell \leq 768$
Resolution	$N_{\rm side} = 1024$, $\ell_{\rm max} = 2048$
Realizations	100 sky $\times$ 4 disjoint masks = 400
Ground truth	100 full-sky, noise-free, unit-weight

Table 1: Run configuration.

3 Transfer functions

The transfer function quantifies why the masked, weighted *filter-then-square* estimator differs from the full-sky *filter-then-square* ground truth for the same sky. For a Gaussian shear field it is calculable from the mask, the weights W_1, W_2, the filter, and the shear power spectrum $C_\ell^{\gamma\gamma}$ alone. We first derive the underlying linear-response model (Section 3.1), which underlies the three *analytic* methods (B)–(D) but *not* the measured (A), and then give four evaluations of T_ℓ, each with its own assumptions: the measured $T^{(\rm A)}$ (Section 3.2), the Gaussian Monte-Carlo $T^{(\rm B)}$ (Section 3.3), the closed form $T^{(\rm C)}$ (Section 3.4), and the mode-coupling-matrix calibration $T^{(\rm D)}$ (Section 3.5). Section 3.6 compares them quantitatively.

3.1 The squeezed-limit response

Filtered-shear kernel. Let $\xi = F[W_1\boldsymbol{\gamma}]$ be the bandpass-filtered (weighted) shear, with real-space bandpass kernel $K(\theta) = \mathrm{bl2beam}(f_\ell)$ (the Legendre transform of the top-hat). Its full-sky, homogeneous two-point function is

$$\rho(\theta) = \mathbb{E}[\xi(\hat{\boldsymbol{n}})\xi^*(\hat{\boldsymbol{n}}')] = \sum_\ell \frac{2\ell+1}{4\pi} f_\ell^2\, C_\ell^{\gamma\gamma}\, P_\ell(\cos\theta), \qquad \mathbb{E}[M_0] = \rho(0), \tag{5}$$

with θ the separation of $\hat{\boldsymbol{n}}, \hat{\boldsymbol{n}}'$ and $C_\ell^{\gamma\gamma} = (C_\ell^{EE} + C_\ell^{BB})/2$ the per-component shear power.

Spin reduction (assumption). ξ is a spin-2 field, so its exact two-point function involves the spin-weighted functions $d^\ell_{2,\pm2}$ (the $\xi_\pm$ correlation functions), not the scalar P_ℓ used in Eq. (5). The scalar series is *exact* at $\theta = 0$, where $d^\ell_{2,2}(0) = P_\ell(1) = 1$ — so the local mean $\mathbb{E}[M_0] = \rho(0)$, and hence the $|K|^2$ noise-bias template, are exact — and is the leading approximation for $\theta > 0$, accurate in the small-separation regime that dominates the narrow-band response. The squared field itself, $M_0 = \xi_1^2 + \xi_2^2 = |\xi|^2$, is a genuine *scalar* (spin-0): the spin-2 structure is integrated out of the modulus, which is why M_0 is crossed with δ as spin-0$\times$spin-0.

Squeezed response. The shear–shear–density signal is the response of the small-scale shear power to the large-scale lens density: across the band $C_\ell^{\gamma\gamma} \to C_\ell^{\gamma\gamma}[1 + b_\phi\,\delta(\hat{\boldsymbol{n}})]$ (a separate-universe / squeezed modulation). To linear order the δ-correlated part of M_0 is the bilinear

$$\delta M_0(\hat{\boldsymbol{n}}) = b_\phi \iint K(\hat{\boldsymbol{n}}, a)K^*(\hat{\boldsymbol{n}}, b)\, W_1(a)W_1(b)\, C^{\gamma\gamma}(a\cdot b)\,\delta(a)\,\mathrm{d}a\,\mathrm{d}b \;+\; (a \leftrightarrow b), \tag{6}$$

where W_1 carries the binary mask. Doing the inner integral, $\int K^*(\hat{\boldsymbol{n}}, b)C^{\gamma\gamma}(a\cdot b)\mathrm{d}b$ has harmonic coefficients $f_\ell C_\ell^{\gamma\gamma}$; treating the two filtered legs as co-located on the kernel scale ($W_1(b) \simeq W_1(a)$, so $W_1(a)W_1(b) \to W_1^2$) gives the compact form

$$\boxed{\delta M_0(\hat{\boldsymbol{n}}) = 2b_\phi\,[\kappa \star (W_1^2\delta)](\hat{\boldsymbol{n}}), \qquad b_\ell^\kappa = \mathrm{beam2bl}\big[\mathrm{bl2beam}(f_\ell)\cdot\mathrm{bl2beam}(f_\ell C_\ell^{\gamma\gamma})\big].} \tag{7}$$

For a white band $C_\ell^{\gamma\gamma} = P_0$ this reduces to $b_\ell^\kappa = P_0\,\mathrm{beam2bl}[\,|K(\theta)|^2]$, i.e. the squared-kernel $|K|^2$ of the optimal-weights note (its noise template, with $n^2 \to P_0$). The shear power enters only through the *shape* of κ (its width within the band); the response amplitude b_ϕ cancels in every transfer below. The co-location step that produced W_1^2 in Eq. (7) from the genuinely bilinear $W_1(a)W_1(b)$ of Eq. (6) is common to the analytic methods (B)–(D), but it returns to bite method (D) (Section 3.5).

Linearity, and why a Gaussian probe suffices. The decisive feature of Eq. (7) is that δM_0 is *linear in* δ. The estimator's signal is the cross of the weighted, masked, decoupled δM_0 with δ; because δM_0 is linear in δ, this cross is a *second moment* (power-spectrum-like) functional of δ, not a three-point function. On the real (mildly non-Gaussian) sky the physical signal $\langle M_0 \delta\rangle$ is a genuine bispectrum — it vanishes identically for a jointly Gaussian shear+density field by Wick's theorem, and is non-zero only because nonlinear growth makes the small-scale shear power track the large-scale density (the amplitude b_ϕ). The transfer, however, asks only how the survey window reshapes the *linear* response Eq. (7); the non-Gaussianity has been folded analytically into the response operator κ and its amplitude b_ϕ, which cancels in the transfer ratio. A probe δ with the correct two-point statistics — in particular a Gaussian draw — is therefore sufficient (and correct) to evaluate the transfer; this is what method (B) exploits.

Transfer. Full sky with $W_1 = 1$ gives the truth response $\delta M_0^{\rm t} = 2b_\phi\, \kappa \star \delta$, so $\langle \delta M_0^{\rm t} \delta\rangle_\ell = 2b_\phi b_\ell^\kappa C_\ell^{\delta\delta}$. NaMaster already removes the cross mode-coupling of the (W_2 mask, mask) weights, so the residual transfer is the field-level window relating the constructed response $\kappa \star (W_1^2\delta)$ to the truth $\kappa \star \delta$:

$$T_\ell = \frac{\big\langle\, \mathrm{decouple}_{W_2 \mathrm{m}}\, \mathrm{pseudo}C_\ell[(W_2\, \mathrm{m})\, \kappa \star (W_1^2 \delta),\ \mathrm{m}\,\delta] \big\rangle}{\big\langle\, \mathrm{decouple}_1\, \mathrm{pseudo}C_\ell[\kappa \star \delta,\ \delta] \big\rangle}, \tag{8}$$

independent of b_ϕ and of the normalization of κ.

Density assumption. We treat $\delta_{\rm src}$ as an externally *fixed survey window*: it sets the depth and hence the noise and the weights W_1, W_2, which are therefore deterministic maps (like the mask). The lens density δ — at a different redshift — is the random probe, taken uncorrelated with the window, $\delta_{\rm src} \perp \delta$. The prediction thus uses only the spectra $C^{\gamma\gamma}$ (via κ) and $C^{\delta\delta}$ (the probe), the mask, the filter and the weights; it never uses the lens map.

Spin-4 (assumption). A scalar power modulation does not source $M_{4E} = \mathrm{Re}\,\xi^2$ at linear order ($\mathbb{E}[\xi^2] = 0$); its signal is the anisotropic (tidal) response, with a different kernel and amplitude. The *geometric* transfer — the mask $\otimes$ filter non-commutation and the weight normalization — is, however, common to both spin channels, so we apply the scalar M_0 prediction to M_{4E} as well. This is confirmed empirically by the near-equal measured M_0 and M_{4E} transfers (Section 3.2, Fig. 2).

3.2 (A) Measured transfer

The measured transfer is the ratio of the (mean) masked, weighted signal bandpower to the full-sky truth,

$$T_\ell^{(\mathrm{A})} = \frac{\langle \hat{C}_\ell^{\rm sig}\rangle}{C_\ell^{\rm truth}}, \tag{9}$$

calibrated on the 400 noise-free signal realizations of the main run — which use the *real*, weakly correlated $\delta_{\rm src}$ and δ, so it makes none of the modelling assumptions of Section 3.1 (no squeezed

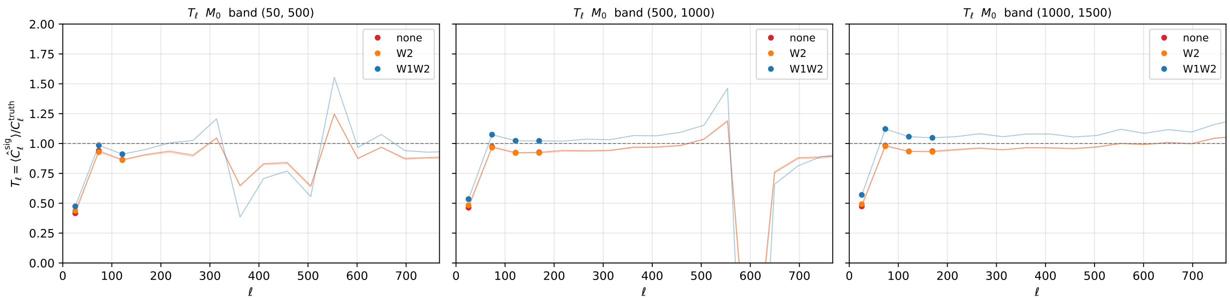

Figure 1: Measured transfer function $T_\ell^{(\mathrm{A})} = \langle \hat{C}_\ell^{\mathrm{sig}} \rangle / C_\ell^{\mathrm{truth}}$ for the M_0 cross, from the noise-free masked realizations, for the three weighting schemes and three filter bands, over the full analysed range $\ell \leq 768$ (faint lines; solid markers flag the signal-bearing bandpowers). **none** and **W2** overlap (NaMaster removes the W_2 weighting from the mean); **W1W2** is shifted upward by the $W_1 = 1/n^2$ modulation, most cleanly in the $[1000, 1500]$ band where T_ℓ is flat across all ℓ. The transient excursion near $\ell \simeq 600$ in $[500, 1000]$ is a zero-crossing of the truth cross-spectrum, where the ratio is ill-defined. The dashed line is unity.

limit, no white-band κ, no fixed-window). It is, however, *circular*: Eq. (9) is constructed so that the corrected signal mean equals the truth by definition, so it cannot test the bias of the signal mean, only the noise-debiasing (Section 4.1). It also requires signal simulations and so is not a first-principles calibration; we use it as the reference against which (B)–(D) are judged.

Figure 1 shows $T_\ell^{(\mathrm{A})}$ for the three schemes (M_0). The **none** and **W2** curves are essentially identical (plateau at $T \simeq 0.93$), confirming that W_2 — supplied as the NaMaster weight — does *not* bias the mean: NaMaster's mode-decoupling removes it. The **W1W2** curve sits higher ($T \simeq 1.0$–1.08 in the noise-dominated bands), the signature of the $W_1 = 1/n^2$ pre-filter modulation. All curves drop in the lowest-ℓ bandpower, where the $\sim 11\%$ footprint poorly recovers the largest scales and the mode-coupling matrix is ill-conditioned.

3.3 (B) Gaussian Monte-Carlo

Method (B) evaluates Eq. (8) by Monte-Carlo, and is the fiducial calibration. The essential point — and the resolution of the apparent paradox that a cubic statistic of a Gaussian field averages to zero — is that (B) does *not* cross a random δ against the data's M_0, nor does it take any three-point average. It replaces M_0 by its *linear response* Eq. (7) and crosses that with the *same* δ, so the quantity averaged is the two-point functional $\langle \delta \cdot \kappa \star (W_1^2 \delta) \rangle$, which is robustly non-zero. Concretely, for each of N_{MC} draws:

1. draw a Gaussian δ from $C_\ell^{\delta\delta}$ (measured from the lens maps);
2. build the model response map $\kappa \star (W_1^2 \delta)$ (a deterministic, *linear* function of the drawn δ);
3. push it through the *identical* NaMaster pipeline used on the data — weighted by W_2 on the masked sky, crossed with $\mathrm{m}\,\delta$, and decoupled with the workspace built from (mask, W_2) — to form the numerator of Eq. (8);
4. form the denominator from the same δ full-sky, unit-weight, $\kappa \star \delta$;
5. accumulate; the ratio of the two ensemble averages is $T_\ell^{(\mathrm{B})}$.

Assumptions. The squeezed-limit linear response and the white-band/scalar-kernel shape of κ (Section 3.1), the fixed-window treatment, and finite-$N_{\rm MC}$ noise. We use $N_{\rm MC} = 500$; convergence is confirmed by comparing with $N_{\rm MC} = 30$, which gives identical χ^2 values to within integer rounding (Table 2). Crucially, the NaMaster decoupling itself is performed *exactly* on the model maps, so (B) captures the full, ℓ-by-ℓ joint mode-coupling of the mask and the weights; it makes no approximation beyond those in the response. The amplitude b_ϕ appears identically in numerator and denominator and cancels; $C_\ell^{\delta\delta}$ appears in both and largely cancels, so $T^{\rm (B)}$ is essentially a geometric window factor and Gaussian δ — having the correct second moment by construction — is the natural, sufficient probe.

3.4 (C) Closed form

Method (C) evaluates the same transfer with no Monte-Carlo, by factorizing it into a mask-edge loss and an inverse-variance Jensen factor,

$$\boxed{T_\ell^{\rm (C)} = g_{\rm edge} \times N_{\rm weight}, \quad g_{\rm edge} = \frac{\langle \kappa \star {\rm m}\rangle_{\rm m}}{\langle {\rm m}\rangle_{\rm m}}, \quad N_{\rm weight} = \frac{\langle W_2\, {\rm m}\, W_1^2\rangle}{\langle W_2\, {\rm m}\rangle}} \tag{10}$$

where $\langle\cdot\rangle_{\rm m}$ is the in-mask average. $g_{\rm edge} \simeq 0.93$–0.94 is the finite-kernel mask-edge loss — the mean fraction of the filter footprint lying inside the mask, set by the filter and $C^{\gamma\gamma}$ — and sets the $\simeq 0.94$ level for the unweighted schemes. $N_{\rm weight}$ is the net inverse-variance Jensen factor: $W_1^2 \propto (1+\delta_{\rm src})^2$ upweights low-noise regions, and although the optimal W_2 partly compensates, a residual enhancement survives, $N_{\rm weight} \approx 1.04$–$1.10$ for W1W2 (rising with band) and $= 1$ for **none**/**W2**.

Additional assumptions (beyond (B)). Relative to (B), the closed form makes three further approximations:

1. *Factorization*: $T_\ell = g_{\rm edge} \times N_{\rm weight}$ treats the edge loss and the weight modulation as independent multiplicative effects, with no cross-talk;
2. *ℓ-independence within a band*: both factors are single numbers per (band, scheme), so $T^{\rm (C)}$ is constant across the cross-multipole bandpowers — it cannot represent the low-ℓ degradation where the coupling matrix is ill-conditioned, which (B) retains;
3. *Mean-field reduction*: both $g_{\rm edge}$ and $N_{\rm weight}$ replace the full NaMaster coupling matrices by scalar position-space averages.

Why (C) is nonetheless robust. $N_{\rm weight}$ is a pure *normalization* (a ratio of means), and is the part of the calibration that is exact: because κ has unit integral, $\langle \kappa \star W_1^2\rangle = \langle W_1^2\rangle$, so the small-scale-weight ambiguity that afflicts the matrix version (Section 3.5) does *not* enter the mean. The ℓ-coupling that (C) discards is exactly where the smooth-weight error would live, and since the cross lives at low ℓ (δ long-ranged) while W_1^2 varies at high ℓ, only the mean amplitude matters. (C) then *divides this scalar out by fiat*, calibrating the mean to the model. The result is cruder than (B) but commits only to the robust quantity, and the two agree (Fig. 2).

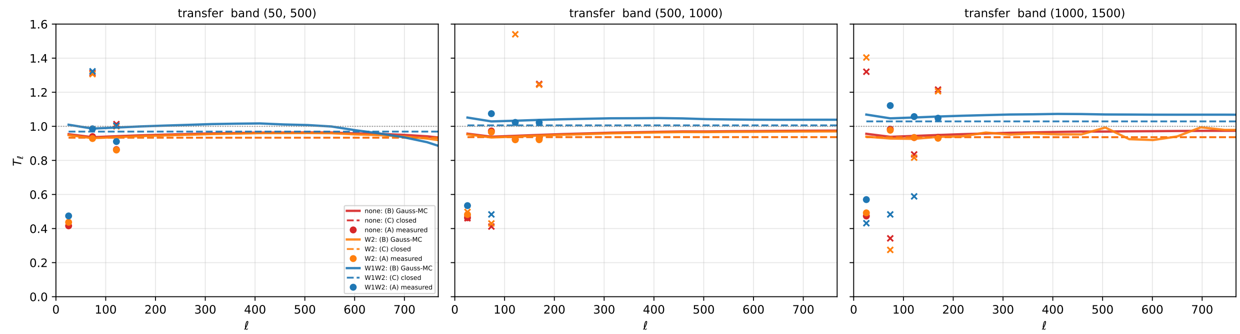

Figure 2: Transfer function by methods (A)/(B)/(C) over the full analysed range $\ell \leq 768$, per scheme (colour) and band. Points: measured (A) for M_0 (circles, signal-bearing bandpowers) and M_{4E} (crosses). Solid lines: Gaussian Monte-Carlo (B); dashed: closed form (C), Eq. (10). The predictions use only the fixed $\delta_{\rm src}$ survey window (hence W_1, W_2), the spectra $C^{\gamma\gamma}$ (via κ) and $C^{\delta\delta}$, the mask and the filter — and never the lens map. (B) and (C) agree, and both match the measured transfer to $\sim$1–3% in the narrow bands $[500, 1000]$ and $[1000, 1500]$ — including the **W1W2** lift to $\simeq 1.0$–1.05 — and to $\sim$8% in the wide low band $[50, 500]$, where the white-band approximation for κ is least accurate (cf. the optimal-weights note). The lift is thus a deterministic inverse-variance effect, not a source–lens correlation; both spin channels follow the common scalar prediction. (The lowest bandpower of the measured T is noisy: the mode-coupling matrix is ill-conditioned for the largest scales on an $\sim$11% footprint.)

3.5 (D) Mode-coupling-matrix calibration

Rather than dividing out a transfer function, method (D) hands the W_1 modulation to NaMaster itself. If the weight were smooth on the kernel scale the response would commute, $\kappa \star (W_1^2 \delta) \simeq W_1^2 (\kappa \star \delta)$, and the weighted squared field entering the pseudo-spectrum would be

$$(W_2 \mathrm{m})\, \delta M_0 \;\simeq\; (W_2\, \mathrm{m}\, W_1^2) \times 2 b_\phi\, (\kappa \star \delta)\,, \tag{11}$$

i.e. the *full-sky truth response* observed through the effective multiplicative window $W_2 \mathrm{m} W_1^2$. Decoupling the measured pseudo-spectrum with the mode-coupling matrix computed for the window pair $(\mathrm{m}, w_{\rm D})$ — in NaMaster, a map-less "dummy" field carrying $w_{\rm D}$ —

$$\boxed{\hat{C}_b^{(\mathrm{D})} = \sum_{b'} \left[M^{(\mathrm{m},\, w_{\rm D})}\right]^{-1}_{bb'} \mathrm{pseudo}C_{b'}\left[(\mathrm{m})\, \delta,\;\, (W_2 \mathrm{m})\, M_0\right]} \tag{12}$$

then estimates the truth response directly: the inverse-variance Jensen factor ($N_{\rm weight}$ of method (C)) is absorbed into the coupling matrix, as a full ℓ-dependent matrix rather than a scalar, and *no transfer function is applied.* The arbitrary W_1 normalization cancels ($M_0 \propto W_1^2$ and $w_{\rm D} \propto W_1^2$), and for **none**/**W2** ($W_1 = 1$) the construction reduces to the standard decoupling.

Choice of window: raw vs kernel-smoothed. The survey weight is *not* smooth on the kernel scale ($\delta_{\rm src}$ has structure at all ℓ). Writing out the δ-contraction of the pseudo-spectrum,

$$\left\langle (\mathrm{m}\, \delta)(x)\;\, (W_2 \mathrm{m})(\hat{\boldsymbol{n}})\, [\kappa \star (W_1^2 \delta)](\hat{\boldsymbol{n}}) \right\rangle = \int \mathrm{m}(x)\, (W_2 \mathrm{m})(\hat{\boldsymbol{n}})\, \kappa(\hat{\boldsymbol{n}} - a)\, W_1^2(a)\, \xi_\delta(x - a)\, \mathrm{d}a, \tag{13}$$

the weight is probed at the *inner* point a of the kernel convolution: since the probe ξ_δ is long-ranged (the cross lives at low ℓ) while κ is narrow, the multiplicative-window model holds with the *kernel-smoothed* weight,

$$w_{\mathrm{D}}^{\mathrm{smooth}} = W_2\,\mathrm{m}\,(\kappa \star W_1^2)\,, \tag{14}$$

(κ normalized to unit integral) rather than with the raw $w_{\mathrm{D}}^{\mathrm{raw}} = W_2 \mathrm{m} W_1^2$ of the naive commutation: the raw window over-counts the small-scale weight structure that the kernel filters out of the band. With $w_{\mathrm{D}}^{\mathrm{raw}}$ the idealized residual is the kernel-edge factor g_{edge} of Eq. (10) (the smoothing of W_1^2, which carries the mask, is not represented); with $w_{\mathrm{D}}^{\mathrm{smooth}}$ the window itself contains the kernel's mask-edge dip — $\kappa \star W_1^2$ falls off where the kernel support leaves the footprint — so the expected residual is *unity*: no transfer function at all.

Implementation. (i) Both decouplings act linearly on the same measured pseudo-spectrum, so the (D) bandpowers follow from the stored standard-decoupled ones exactly, $\hat{C}^{(\mathrm{D})} = M_{\mathrm{D}}^{-1} M_{\mathrm{std}}\,\hat{C}^{\mathrm{std}} \equiv A\,\hat{C}^{\mathrm{std}}$ (binned matrices), with no re-measurement; we verified this re-decoupling against a direct (D) decoupling of one realization to relative precision $\lesssim 4 \times 10^{-15}$. (ii) The coupling matrices use the fixed survey window of realization 1 — the same fixed-window treatment as (B)/(C) — while the measured M_0 carries the per-realization W_1. (The *ensemble-mean* noise map must not be used here: averaging over source-density realizations washes out the modulation, leaving a nearly flat window and a no-op correction.) (iii) For the spin-4 channel the decoupling in principle mixes (E, B); we apply the $E \to E$ block of the re-decoupling matrix to the stored E bandpowers, having checked that the $E \leftarrow B$ leakage block vanishes identically for these windows.

The extra approximation, and why (D) can do worse than (C). Method (D) is often described as the matrix generalization of (C), but it is more precisely a matrix built from an *approximate* effective-window model — promoting (C) exactly to a matrix would merely reproduce (B). The response Eq. (7) is a *convolution* of the weight with the field; to use it as a decoupling matrix, (D) must rewrite it as a *single multiplicative window* $w_{\mathrm{D}}(\hat{\boldsymbol{n}})$ times the truth response $\kappa \star \delta$. No single-point window can do this exactly, because the underlying response Eq. (6) is genuinely *bilinear* in the weight, carrying W_1 at *two* filtered points a, b: weight fluctuations above the field bandpass cannot modulate the band at all (they are filtered out at both legs), whereas $\kappa \star W_1^2$ still credits their positive contribution to the local mean square. The kernel-smoothed window Eq. (14) fixes the leading term but leaves this residual, so the matrix *over-corrects* slightly, leaving the recovered bandpower below truth.

(C) is immune to this because it never represents the operator: it extracts only the global amplitude, a ratio of means that is exact (Section 3.4), and divides it out by fiat. (B) is immune because it never factorizes at all — it decouples the actual response map through the exact matrix. (D), by contrast, commits to a full ℓ-coupling derived from a structurally wrong window *and* applies no compensating transfer, so the residual model error is left in. The error is worst where the kernel is broadest — the wide band $[50, 500]$ — where the commutation is poorest and the white-band κ approximation already degrades (B) and (C); in the narrow bands the kernel is tight and (D) $\approx$ (B) $\approx$ (C).

Figure 3 tests both windows on the signal-only channel (M_0). The raw window *over*-corrects: the residual transfer lands below g_{edge} (medians 0.92 in $[1000, 1500]$ and 0.89 in $[500, 1000]$, against $g_{\mathrm{edge}} \simeq 0.94$), because the coupling matrix charges the estimator for small-scale weight power the kernel has already filtered out. The kernel-smoothed window behaves as derived: its residual is consistent with *unity* to 2% in $[1000, 1500]$ ($\langle T \rangle = 0.98$) and to 5% in $[500, 1000]$ ($\langle T \rangle = 0.95$) — the W_1 lift *and* the edge loss both absorbed by the matrix, no transfer applied. (The two variants

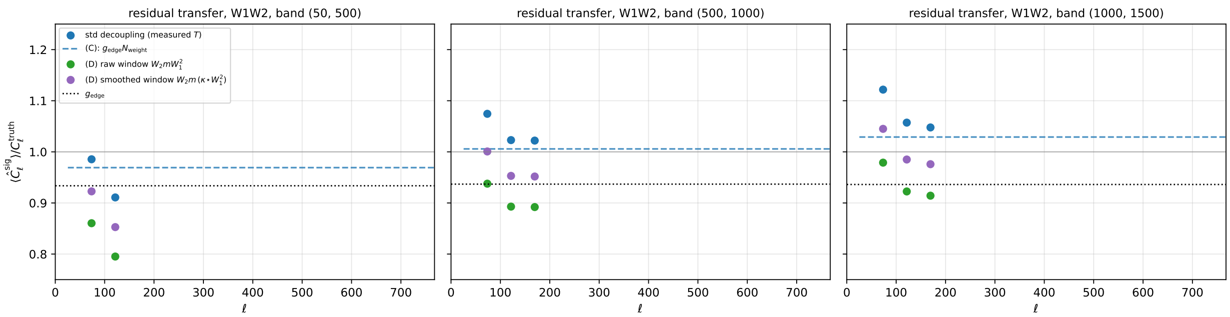

Figure 3: Residual transfer of the **W1W2** scheme under the standard decoupling (blue points: measured; blue dashed: the closed form $g_{\rm edge}N_{\rm weight}$ of Eq. (10)) and under the method-(D) matrix decoupling, with the raw window $W_2{\rm m}W_1^2$ (green) and the kernel-smoothed window $W_2{\rm m}(\kappa\star W_1^2)$ of Eq. (14) (purple), for the M_0 channel per filter band, on the signal-bearing bandpowers. The dotted line is $g_{\rm edge}$, the solid grey line unity. The raw window over-corrects (lands below $g_{\rm edge}$); the kernel-smoothed window absorbs both the W_1 lift and the edge loss, leaving a residual close to unity in the narrow bands.

differ by a nearly flat factor $\simeq 1/g_{\rm edge}$.) In the wide band $[50, 500]$ the deficit grows to $\sim 15\%$. The spin-4 channel follows the same pattern within its much larger transfer uncertainty (median smoothed-window residuals 0.87–0.90).

3.6 Comparison

Table 2 collects the truth-residual χ^2 of the fiducial **W1W2** M_0 bandpowers under each calibration. The χ^2 is diagonal, over the $\nu = 16$ analysed bandpowers ($\ell \leq 768$), and uses the 400-realization error *on the mean*; its statistical expectation is $\nu = 16$ with spread $\pm\sqrt{2\nu} \approx 5.7$. Values well above ν therefore flag a residual *model bias* (not scatter), amplified by the small error on the mean. To make that bias intuitive, the parenthetical gives the corresponding χ^2 *excess per single realization*, $(\chi^2 - \nu)/N_{\rm real}$ with $N_{\rm real} = 400$: it is the systematic contribution as a single realization would see it (negative values mean the residual is below the statistical expectation, i.e. no detectable bias). All four calibrations leave a per-realization excess $\ll 1$ — equivalently $\lesssim 0.15\sigma$ per bandpower.

The ordering is the one the derivations predict. (A) is unbiased by construction and sits at or below ν. (B) and (C) — the Monte-Carlo and closed-form evaluations of the *same* scalar transfer — are comparable, with the wide band $[50, 500]$ the worst for both (the white-band κ approximation). (D), which keeps the full ℓ-coupling but pays for it with the single-window approximation and applies no compensating transfer, matches (B)/(C) in the narrow bands and is the worst of the four in the wide band ($\chi^2 = 96$). On this smooth footprint (D) is thus *not* more accurate than (B); its distinct value is being simultaneously Monte-Carlo–free, free of any explicit transfer function, and matrix-level (it corrects bandpower-to-bandpower couplings of the weight, which the scalar (C) cannot). The fiducial analysis adopts (B). MC convergence of (B) has been verified: repeating the Gaussian draw with $N_{\rm MC} = 500$ gives χ^2 values identical to $N_{\rm MC} = 30$ to within integer rounding, confirming that 30 draws are already well into the converged regime for this geometry.

The M_{4E} rows test the spin-independence assumption of Section 3.1: all three analytic methods reuse the M_0 geometric transfer for the spin-4 channel. They calibrate it comparably in the narrow bands, but visibly under-perform in the signal-dominated wide band $[50, 500]$ — where (D) reaches $\chi^2 = 320$ — because the spin-4 transfer genuinely differs there (the tidal response has its own kernel)

method	field	$[50, 500]$	$[500, 1000]$	$[1000, 1500]$
(A) measured†	M_0	4.0 (-0.03)	6.1 (-0.02)	12.2 (-0.01)
	M_{4E}	9.1 (-0.02)	9.1 (-0.02)	21.0 (0.01)
	M_{4B}	28.7 (0.03)	7.5 (-0.02)	16.4 (0.00)
(B) Gaussian MC *(fid.)*	M_0	56 (0.10)	34 (0.05)	14 (0.00)
	M_{4E}	78 (0.15)	20 (0.01)	40 (0.06)
	M_{4B}	28.7 (0.03)	7.5 (-0.02)	16.4 (0.00)
(C) closed form	M_0	46 (0.08)	42 (0.07)	18 (0.01)
	M_{4E}	50 (0.09)	14 (-0.01)	34 (0.05)
	M_{4B}	28.7 (0.03)	7.5 (-0.02)	16.4 (0.00)
(D) matrix (smoothed)	M_0	96 (0.20)	49 (0.08)	17.5 (0.00)
	M_{4E}	320 (0.76)	29 (0.03)	41 (0.06)
	M_{4B}	28.7 (0.03)	7.5 (-0.02)	16.4 (0.00)

Table 2: Truth-residual χ^2 ($\nu = 16$, diagonal, 400-realization error on the mean) of the **W1W2** M_0, spin-4 M_{4E}, and parity null-test M_{4B} cross-spectra after each transfer calibration, per filter band. In parentheses: the χ^2 excess per single realization, $(\chi^2 - \nu)/400$. †(A) is circular — calibrated *per channel* on the same signal sims, so its mean is unbiased by construction (excess ≤ 0); it is the reference, not a first-principles calibration. For M_0, (B)/(C) are comparable and first-principles and (D) matches them except in the wide band $[50, 500]$, where its single-window approximation (Section 3.5) is poorest. The analytic transfers (B)–(D) are the M_0 *(geometric) prediction applied to M_{4E} as well*: they hold comparably in the narrow bands but leave a larger residual in the signal-dominated wide band (most starkly (D), $\chi^2 = 320$), where the spin-4 channel's genuinely different transfer (Section 3.1) matters most and the small M_{4E} error bars resolve the mismatch. M_{4B} is a parity null test (truth $\equiv 0$, no transfer applied); its χ^2 measures deviation from zero. Every per-realization excess is < 1 ($\lesssim 0.2\sigma$ per bandpower).

and the small M_{4E} error bars resolve the mismatch that the field-specific measured (A) absorbs by construction. The per-realization excess nonetheless stays < 1 throughout, so the residual is sub-σ at the single-realization level even for M_{4E}; a dedicated spin-4 response kernel would be needed to calibrate that band analytically.

4 Results

4.1 Unbiasedness

We quantify unbiasedness two ways. First, a clean test of the noise debiasing: per realization, $d_b = \hat{C}_b^{\rm tot} - \hat{C}_b^{\rm sig}$ isolates the noise contribution (the signal cancels sample by sample), and its mean must vanish. Second, the residual of the corrected mean against the truth, $\hat{C}_b^{\rm corr} - C_b^{\rm truth}$, compared with the error on the mean. Both give diagonal χ^2 consistent with the number of bandpowers (Table 3). The transfer functions $T^{(\rm A)}$–$T^{(\rm D)}$ used to put the bandpowers in truth units are derived and compared in Section 3; the fiducial measurement is **W1W2** corrected by the analytic $T^{(\rm B)}$.

Figures 4 and 5 show the cross-spectra corrected by the measured transfer $T^{(\rm A)}$: all three schemes track the full-sky ground truth, and the residual panels scatter at the ± 1 level about zero. Because $T^{(\rm A)}$ is calibrated on the same signal sims, this comparison is partly circular. The fiducial

field	band / scheme		$\langle T\rangle$	$\chi^2_{\mathrm{noise}}/\nu$	$\chi^2_{\mathrm{truth}}/\nu$	SNR
M_0	$[50, 500]$	none	0.88	8.3	5.9	2.9
		W2	0.88	6.1	3.4	3.2
		W1W2	0.95	8.6	4.0	3.2
	$[500, 1000]$	none	0.94	15.0	14.5	2.3
		W2	0.94	7.7	6.9	4.2
		W1W2	1.03	8.4	6.1	**5.0**
	$[1000, 1500]$	none	0.97	13.9	13.5	1.9
		W2	0.96	9.1	8.5	4.1
		W1W2	1.08	15.9	12.2	**5.6**
M_{4E}	$[50, 500]$	none	0.90	18.6	12.1	4.7
		W2	0.89	18.0	10.6	5.0
		W1W2	0.93	18.1	9.1	5.0
	$[500, 1000]$	none	0.92	12.2	11.9	0.8
		W2	0.92	8.7	7.7	1.3
		W1W2	0.94	10.6	9.1	1.4
	$[1000, 1500]$	none	0.95	13.3	13.2	0.5
		W2	0.95	13.0	12.4	1.1
		W1W2	0.95	22.5	21.0	1.2

Table 3: Transfer-function median $\langle T\rangle$ (method A), noise-debiasing χ^2, truth-residual χ^2 (both diagonal, $\nu = 16$ bandpowers on $\ell \leq 768$), and detection SNR, per field / filter band / weighting scheme, from the 400-realization NSIDE$= 1024$ run. The χ^2 values are all consistent with $\nu = 16$ (expected spread $\pm\sqrt{2\nu} \approx 5.7$). The SNR rises sharply with weighting in the noise-dominated bands (bold).

result, Figs. 6 and 7, instead divides by the *independent* analytic transfer $T^{(\mathrm{B})}$ (the M_0 prediction, used for both channels since the geometric transfer is spin-independent). The bandpowers remain consistent with truth — residuals at the ±1–2 level in the narrow bands $[500, 1000]$ and $[1000, 1500]$, with a mild trend in the wide band $[50, 500]$ reflecting the $\sim 8\%$ white-band inaccuracy of $T^{(\mathrm{B})}$ there. The estimator is thus unbiased even under a fully first-principles calibration. Figures 8 and 9 repeat the comparison with the Monte-Carlo–free closed-form transfer $T^{(\mathrm{C})}$; the bandpowers are indistinguishable from the $T^{(\mathrm{B})}$ result. Finally, Figs. 10 and 11 show method (D), in which no transfer function is applied to the weighted scheme at all: the matrix-decoupled **W1W2** bandpowers (**none**/**W2** corrected by $T^{(\mathrm{C})} = g_{\mathrm{edge}}$ as before) track the truth to 2% in $[1000, 1500]$ and 5% in $[500, 1000]$ in the mean over the 400 realizations, degrading in the wide band $[50, 500]$ as discussed in Section 3.5. The per-realization significance of every residual is $\lesssim 0.2\sigma$ (Table 2).

4.2 Covariance cross-check

Because the four footprints are equal-area and disjoint, the covariance of one sample equals the NaMaster Gaussian (Knox-like) covariance for a single footprint, which we evaluate from the on-the-fly $\delta\delta$, M_0M_0, δM_0 auto/cross spectra. The diagonal of the Gaussian covariance matches the 400-sample scatter (Table 4), validating the analytic error bars.

Figure 12 compares the two error estimates directly: the per-realization $\sigma(\hat{C}_\ell^{\delta M_0})$ from the 400-realization scatter (points) and from the NaMaster Gaussian covariance (lines) lie on top of each other across all bands and schemes, except the single anomalous W2, $[1000, 1500]$ configuration (the high orange line in the right panel).

Figure 13 shows the correlation matrices of the 400-realization covariance: they are strongly

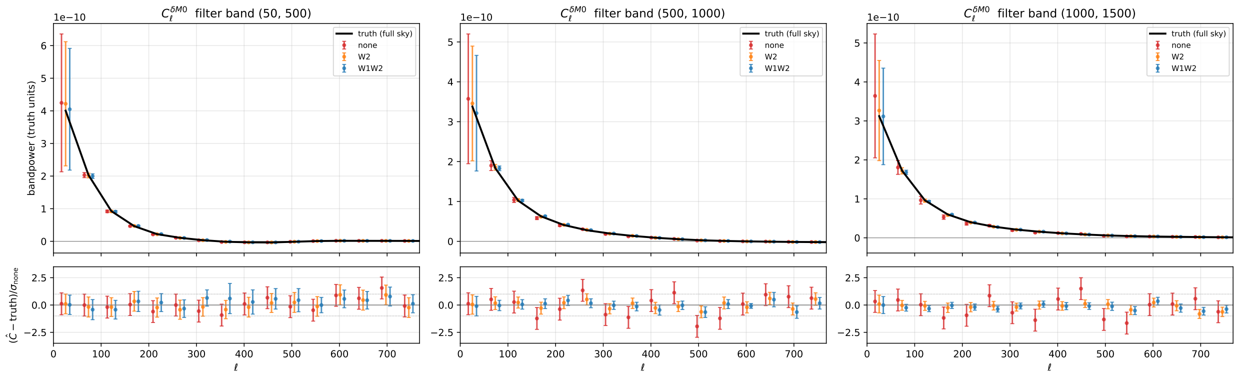

Figure 4: $\hat{C}_\ell^{\delta M_0}$ in truth units for the three weighting schemes (points, offset for clarity) against the full-sky ground truth (black), per filter band, corrected by the *measured* transfer $T^{(\mathrm{A})}$. Points are the *mean over the* 400 *realizations*; error bars are the error on that mean, $\sqrt{\mathrm{diag}\, C/400}$, with C the sample covariance from the scatter between the 400 realizations. *Lower panels:* residual $(\langle \hat{C} \rangle - \mathrm{truth})/\sigma_{\mathrm{none}}$, in units of the **none** error on the mean; points scatter at the ± 1 level and are consistent with zero (unbiased), while the shrinking bars from **none**→**W2**→**W1W2** (fiducial) in the noise-dominated bands are the SNR gain.

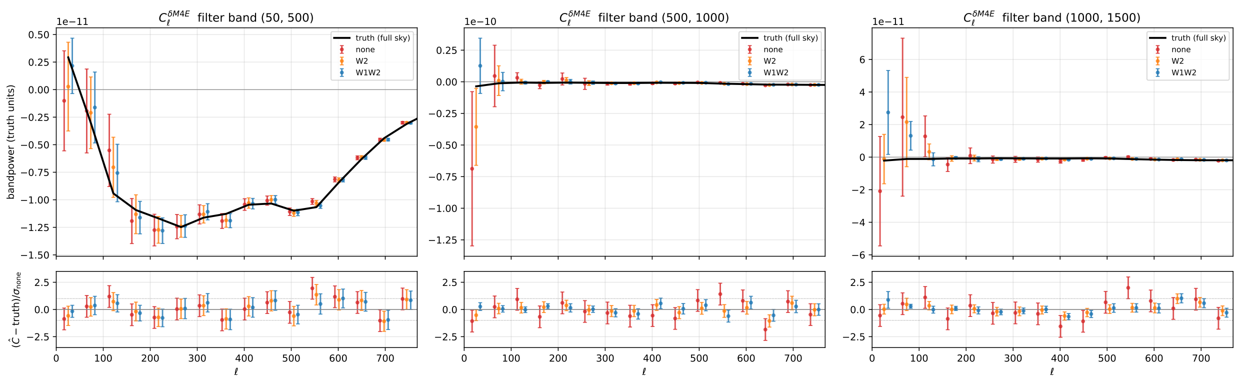

Figure 5: As Fig. 4 for the spin-4 E-mode cross $\hat{C}_\ell^{\delta M_{4E}}$ (no noise bias), corrected by $T^{(\mathrm{A})}$.

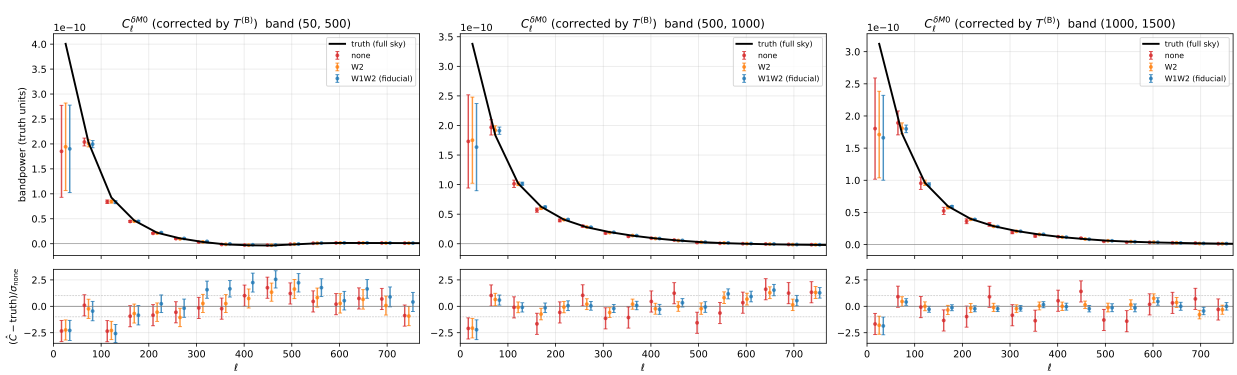

Figure 6: **Fiducial.** $\hat{C}_\ell^{\delta M_0}$ as Fig. 4 but corrected by the *independent* analytic transfer $T^{(\mathrm{B})}$ (predicted from the survey window and power spectra, no signal-sim input).

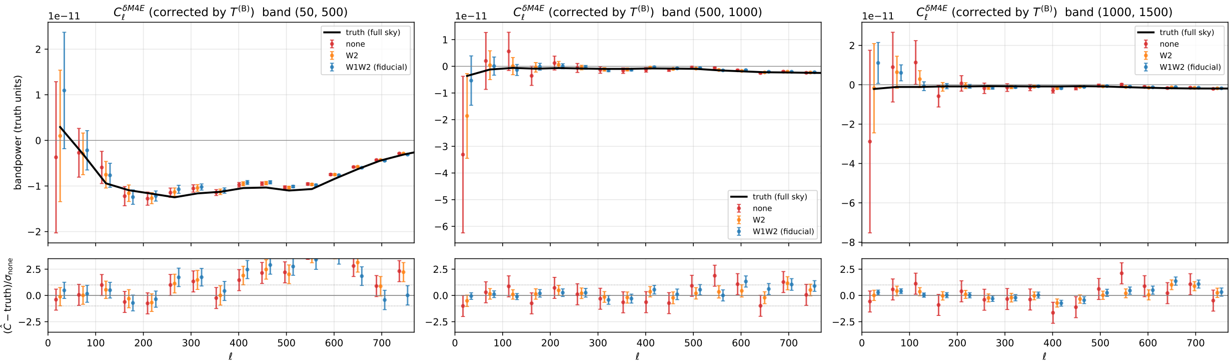

Figure 7: As Fig. 6 for the spin-4 cross $\hat{C}_\ell^{\delta M_{4E}}$, corrected by the analytic $T^{(\mathrm{B})}$.

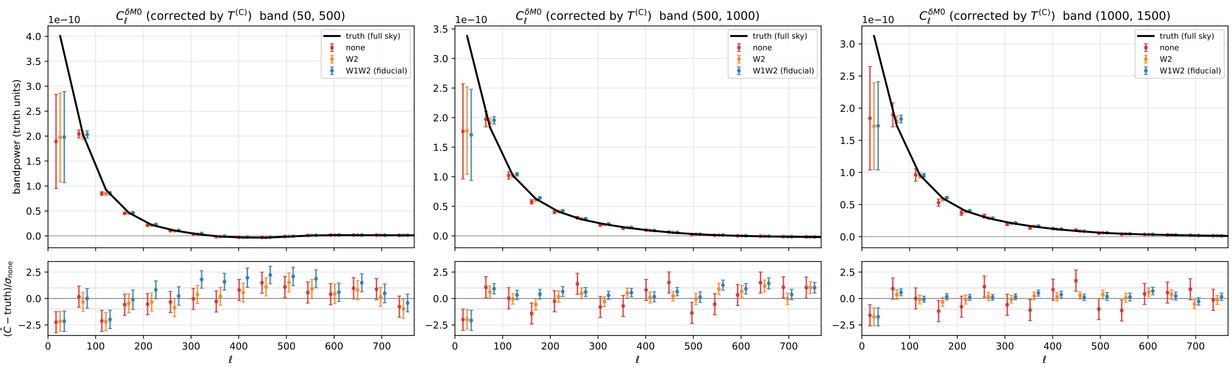

Figure 8: As Fig. 6 but corrected by the *closed-form* analytic transfer $T^{(\mathrm{C})}$ of Eq. (10) (the Monte-Carlo–free $g_{\mathrm{edge}} \times N_{\mathrm{weight}}$ factorization, constant within each band). The bandpowers remain consistent with truth, matching the (B) result.

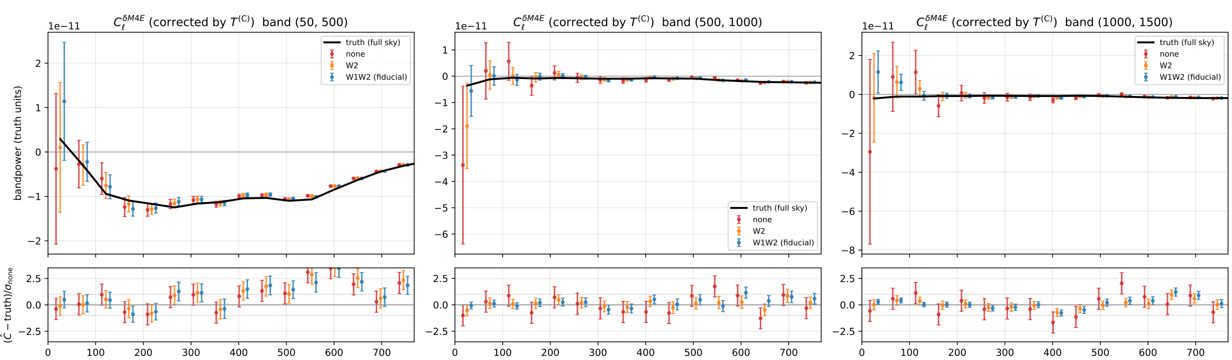

Figure 9: As Fig. 8 for the spin-4 cross $\hat{C}_\ell^{\delta M_{4E}}$, corrected by the closed-form $T^{(\mathrm{C})}$.

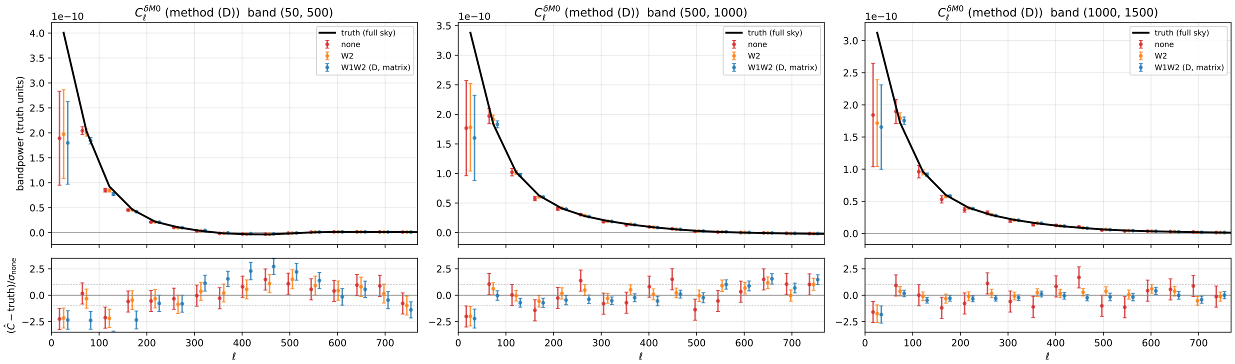

Figure 10: As Fig. 6 but with the **W1W2** channel calibrated by method (D), Eq. (12): the bandpowers are decoupled with the mode-coupling matrix of the kernel-smoothed effective window $W_2\mathrm{m}\,(\kappa \star W_1^2)$ of Eq. (14), which absorbs the W_1 modulation and the edge loss as a matrix — *no transfer function is applied*. The **none**/**W2** schemes (for which (D) reduces to the standard decoupling) are corrected by $T^{(\mathrm{C})} = g_{\mathrm{edge}}$ as before.

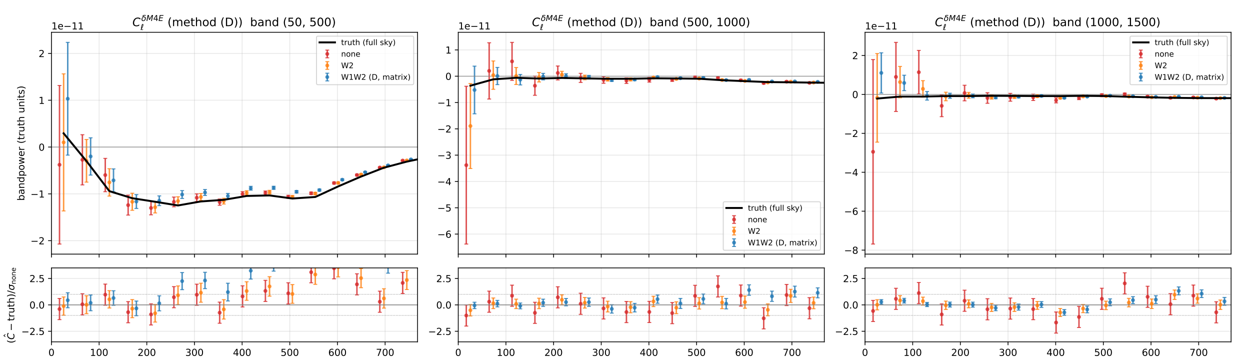

Figure 11: As Fig. 10 for the spin-4 cross $\hat{C}_\ell^{\delta M_{4E}}$, with the **W1W2** channel matrix-decoupled per method (D).

band	none	W2	W1W2
$[50, 500]$	0.97	0.96	0.98
$[500, 1000]$	0.93	0.96	1.00
$[1000, 1500]$	0.95	$268^{\dagger}$	1.00

Table 4: Median ratio $\sqrt{\operatorname{diag} C_{\rm Gauss}}/\sqrt{\operatorname{diag} C_{\rm sample}}$ of the NaMaster Gaussian error to the 400-sample scatter for the M_0 cross. The agreement is at the few-percent level except for one configuration (†): the unweighted-shear, highest-band W2 weight $\propto 1/(|K|^2 \star n^2)^2$ is extremely peaked (the band-$[1000, 1500]$ noise spans more than a decade), which destabilises the on-the-fly Gaussian-covariance inputs; the W1W2 scheme, which pre-whitens with $W_1 = 1/n^2$, is stable there (1.00). This affects only the analytic cross-check, not the sample-covariance results.

diagonal-dominant for both M_0 and M_{4E} in all bands (off-diagonals $|r| \lesssim 0.25$), with only a mild anti-correlation between the two lowest-ℓ bandpowers (the ill-conditioned largest scales on the $\sim 11\%$ footprint). The near-diagonal covariance justifies the diagonal χ^2 used above.

4.3 Signal-to-noise and the gain from weighting

In the low band $[50, 500]$ the cross variance is signal-dominated (the shape noise contributes $\sim 1\%$ of $\mathrm{Var}[\hat{C}_\ell^{\delta M_0}]$), so optimal weighting yields no gain — consistent with the note, whose weights minimize the noise-dominated term. In the higher, noise-dominated bands the inverse variance weighting reduces the error: Fig. 14 compares the per-bandpower error of the three schemes, and the detection SNR in Table 3 rises substantially from **none** to **W2** to **W1W2**: for M_0, $2.3 \to 4.2 \to 5.0$ in $[500, 1000]$ and $1.9 \to 4.1 \to 5.6$ in $[1000, 1500]$ — a factor of 2.2–2.9 improvement in significance, of which most comes from the post-square weight W_2 and a further $\sim$ 20–35% from adding the pre-filter W_1. The spin-4 M_{4E} cross shows the same pattern ($0.5 \to 1.2$ in the top band). In the signal-dominated band $[50, 500]$ the gain is marginal ($2.9 \to 3.2$), as expected.

5 Code overview

The estimator is packaged as the installable Python module `fsb_ssg` (*Filtered-Square Bispectrum: shear–shear–galaxy*); this validation lives in `science/` of the same repository and runs with pymaster 2.7 and healpy 1.19. The generic estimator is in `src/fsb_ssg/`; the survey-specific adapter `science/pipeline/simlib.py` binds it to the simulation configuration and map IO, so the physics lives in one place. The pipeline is staged by `science/pipeline/run_pipeline.sh` (`FSB_NSIDE`, `FSB_SIM_DIR`; masks run in parallel at $N_{\rm side} \geq 1024$):

`fsb_ssg` **(library)** core library: the inhomogeneous noise model, W_1/W_2 construction (Eq. (4)), the filtered-square step, the `bl2beam` $|K|^2$ noise-bias template, the squeezed-response kernel κ (Eq. (7)), and the NaMaster spin-0×spin-0 (M_0) and spin-0×spin-4 (M_{4E}) decoupled cross-spectra. Unit-tested on synthetic maps via GitHub Actions; cached map IO is provided by the `simlib` adapter.

`build_cache.py` degrade and cache the input maps to the working resolution (skipped at $N_{\rm side} = 1024$, where the maps are native).

`ground_truth.py` full-sky, noise-free, unit-weight ground truth (through NaMaster with a unit mask, apples-to-apples).

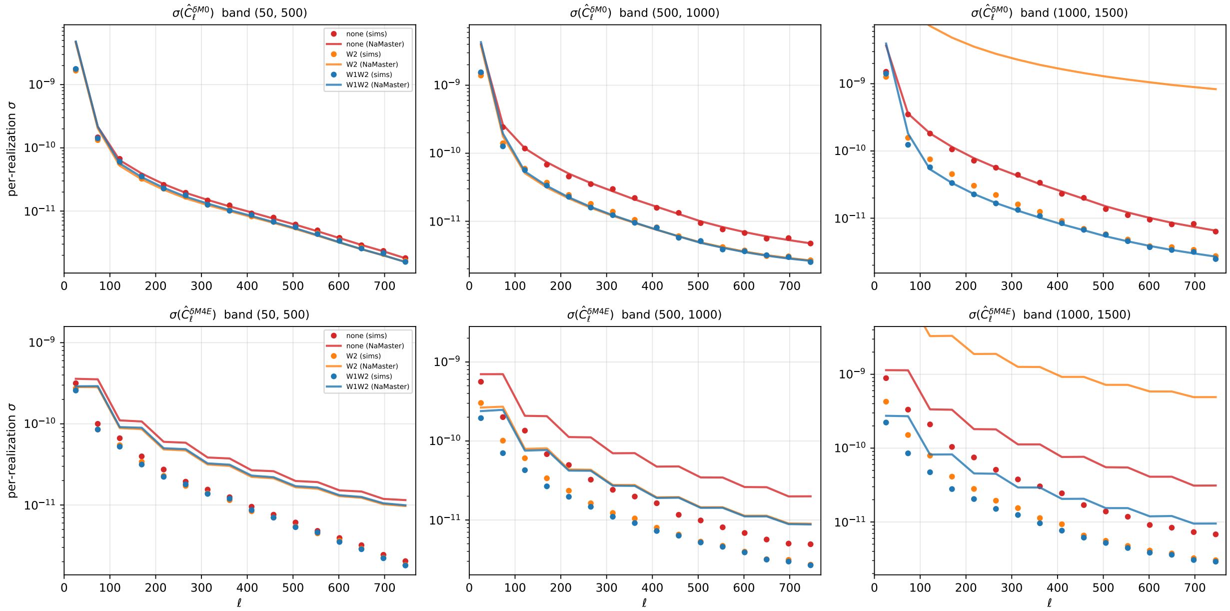

Figure 12: Per-realization error $\sigma = \sqrt{\operatorname{diag} C}$ from the 400-realization sample scatter (points) versus the NaMaster Gaussian covariance (lines), for the three schemes and three bands, for M_0 (top) and M_{4E} (bottom). For the spin-0 M_0 cross the two agree to a few percent (cf. Table 4; the lone outlier is the orange line in $[1000, 1500]$, the peaked unwhitened-W2 weight). For the spin-4 M_{4E} cross the Gaussian estimate runs parallel to the sample σ (the ℓ-shape and scheme ordering match) but high by a near-constant factor ≈ 3: the $f_{\rm sky} = \langle w^2 \rangle$ deconvolution of the input auto-spectra is cruder for spin-weighted fields, whose mask coupling loses more power than the scalar $\langle w^2 \rangle$ captures. In both channels the error drops from **none** (red) through **W2** (orange) to **W1W2** (blue) in the noise-dominated bands — the SNR gain of Table 3. The sample covariance (which is what the analysis uses) is robust for both.

`run_masked.py` + `combine_masks.py` the 100×4 masked, noisy run for the three schemes and the signal-only channel (for $T^{(\mathrm{A})}$); parallel per mask, then combined.

`analyze.py` the two unbiasedness χ^2 tests, the $T^{(\mathrm{A})}$-corrected cross-spectra, the SNR comparison, the correlation matrices, and the table data.

`predict_transfer.py` the analytic transfer (Section 3): methods (B) Gaussian-MC and (C) closed form, the three-method comparison, and the fiducial $T^{(\mathrm{B})}$-corrected cross-spectra.

`nmt_covariance.py` the NaMaster Gaussian covariance versus the sample scatter, for M_0 and M_{4E}.

`method_d.py` the method-(D) mode-coupling calibration (Eq. (12)): the dummy-window ($W_2 \mathrm{m} W_1^2$ and $W_2 \mathrm{m}(\kappa \star W_1^2)$) coupling matrices, the exact linear re-decoupling of the stored bandpowers, its validation, and the (D) figures.

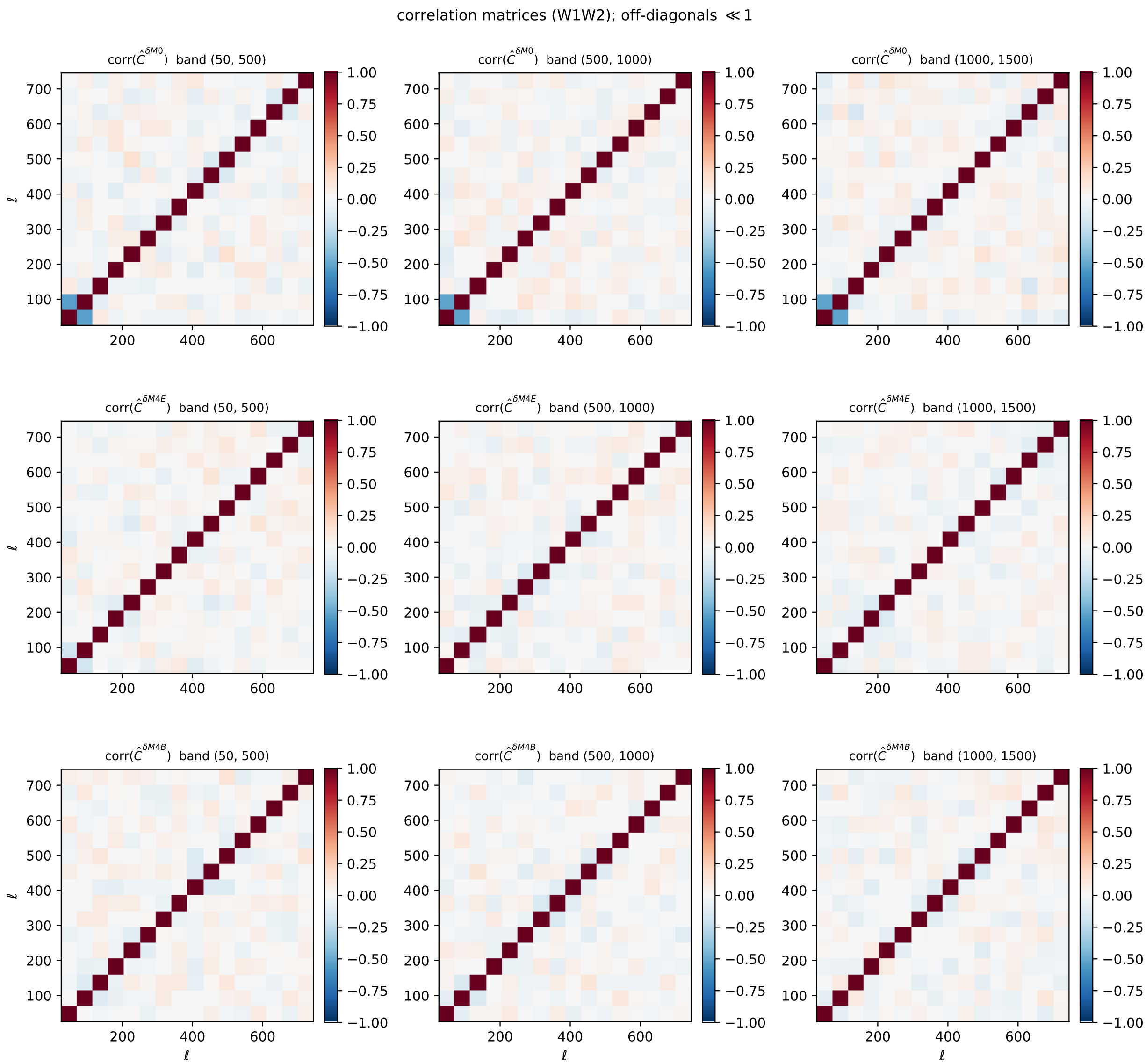

Figure 13: Correlation matrices $C_{\ell\ell'}/\sqrt{C_{\ell\ell}C_{\ell'\ell'}}$ of the 400-realization covariance for the W1W2 scheme, for M_0 (top) and M_{4E} (bottom) in the three bands. The matrices are nearly diagonal; the only visible off-diagonal feature is a mild anti-correlation of the two largest-scale bandpowers.

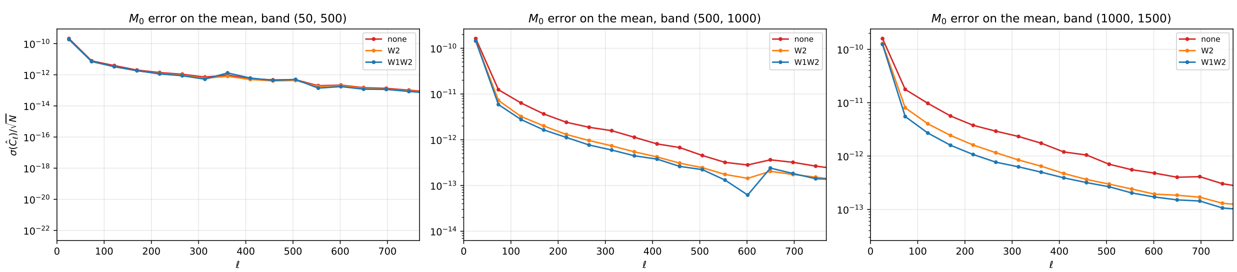

Figure 14: Per-bandpower error on the mean of $\hat{C}_\ell^{\delta M_0}$ for the three schemes. In the noise-dominated bands, optimal weighting (W2, W1W2) lowers the error relative to the unweighted (none) estimator.